

MISSOURI



MISSOURI UNIVERSITY OF SCIENCE AND TECHNOLOGY

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Section 1.4

The Matrix Equation $A\mathbf{x} = \mathbf{b}$

Matrix-Vector Products

If A is an $m \times n$ matrix and if vector $\mathbf{x}$ is in $\mathbb{R}^n$, then the product $A\mathbf{x}$ is the linear combination of the columns of A using the corresponding entries in $\mathbf{x}$ as weights.

Symbolically,

$$A\mathbf{x} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$= x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n$$

Note that the product yields a vector in $\mathbb{R}^m$.

Example 1

Compute the product.

$$\begin{bmatrix} 1 & -3 & 5 & -7 \\ 2 & 4 & -6 & 8 \end{bmatrix} \begin{bmatrix} 2 \\ -3 \\ 1 \\ 4 \end{bmatrix}$$

Matrix-Vector Products

The Row-Vector Rule:

If the product $A\mathbf{x}$ is defined, then the i th entry of the resulting vector is the sum of the products of the corresponding entries from row i of A and the vector $\mathbf{x}$.

Example 2

Consider the system of linear equations

$$x_1 + 2x_2 = -5$$

$$3x_1 + 4x_2 = 1$$

Rewrite this system as a vector equation.

Then, rewrite the linear combination on the left side as a matrix-vector product.

The Matrix Equation $A\mathbf{x} = \mathbf{b}$

If A is an $m \times n$ matrix with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$ and if $\mathbf{b}$ is in $\mathbb{R}^m$, then the matrix equation

$$A\mathbf{x} = \mathbf{b}$$

has the same solution set as the vector equation

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \dots + x_n\mathbf{a}_n = \mathbf{b}$$

and the system of linear equations whose augmented matrix is

$$[\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n \quad \mathbf{b}]$$

When Is $A\mathbf{x} = \mathbf{b}$ Consistent?

The equation $A\mathbf{x} = \mathbf{b}$ has a solution if and only if $\mathbf{b}$ is a linear combination of the columns of A .

It is more challenging (but often more interesting and useful) to answer the following question:

Is $A\mathbf{x} = \mathbf{b}$ consistent for all possible $\mathbf{b}$?

Example 3

$$\text{Let } A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

Is the equation $A\mathbf{x} = \mathbf{b}$ consistent for all possible $\mathbf{b}$?

Example 4

$$\text{Let } A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

Is the equation $A\mathbf{x} = \mathbf{b}$ consistent for all possible $\mathbf{b}$?

$$\begin{bmatrix} 1 & 2 & 3 & b_1 \\ 4 & 5 & 6 & b_2 \\ 7 & 8 & 10 & b_3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 & b_1 \\ 0 & -3 & -6 & b_2 - 4b_1 \\ 0 & 0 & 1 & b_1 - 2b_2 + b_3 \end{bmatrix}$$

Theorem

Let A be an $m \times n$ matrix. Then, the following statements are logically equivalent.

- For each $\mathbf{b}$ in $\mathbb{R}^m$, the equation $A\mathbf{x} = \mathbf{b}$ has a solution.
- Each $\mathbf{b}$ in $\mathbb{R}^m$ is a linear combination of the columns of A .
- The columns of A span $\mathbb{R}^m$.
- A has a pivot position in every row.

Example 5

Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$. Can one column of A be written as a linear combination of the other two?

Example 6

Determine whether every vector in $\mathbb{R}^4$ can be written as a linear combination of the columns of the matrix

$$B = \begin{bmatrix} 1 & 4 & 1 & 2 \\ 0 & 1 & 3 & -4 \\ 0 & 2 & 6 & 7 \\ 2 & 9 & 5 & -7 \end{bmatrix}$$
